

CORRECTION (2026-06-24, DEEP RE-VERIFICATION AT ATIF'S REQUEST)

The first marking of this paper scored 57.5/65 (88.5%, A). A page-by-page re-check against the answer key corrected three errors: Q3(iv) (a correct first attempt was overlooked; the reversed redo was the only part marked), Q2(iii) (a 0.5-mark slip was deducted a full mark), and a Section B addition error (29.5 written as 28.5). Benefit of the doubt was also extended on three borderline items (Q2(ix)OR, Q2(xi)OR, Q3(ii)). Corrected score: 62.5/65 (96.2%, A+).

Chemistry Exam 12 — Chapters 1-9 (Full Syllabus)

Seemab Ali • Class 9 • Federal Board (FBISE)

Attempt date: 2026-06-24 • Marked: 2026-06-24 (corrected same day)

SECTION A (MCQS)

12/12

100%

SECTION B (SHORT)

31.5/33

95.5%

SECTION C (LONG)

19/20

95%

TOTAL SCORE

62.5/65

96.2%

A+

Section A — Multiple Choice Questions (12/12)

Q	Correct Answer	Seemab's Answer	Result	Marks
Q1	A	A	✓	1
Q2	B	B	✓	1
Q3	C	C	✓	1
Q4	A	A	✓	1
Q5	D	D	✓	1
Q6	B	B	✓	1
Q7	C	C	✓	1
Q8	A	A	✓	1
Q9	D	D	✓	1
Q10	B	B	✓	1
Q11	C	C	✓	1
Q12	A	A	✓	1

Perfect score. Q9 and Q10 were self-corrected on the answer sheet; both are correct and credited in full. This is Seemab's 5th consecutive perfect-MCQ Chemistry paper and ties the overall MCQ record.

Section B — Short Answers (31.5/33)

Q2(i) — MAIN *Definition of chemistry + 3 branches*

3/3

Full marks. Chemistry correctly defined. Organic, inorganic, and biochemistry each described accurately with examples. No deductions.

Q2(ii) — OR *Solution / suspension / colloid comparison table*

2.5/3

Table structure good; particle sizes and settling behaviour all correct.

-0.5: Suspension described as "homogeneous" — it is heterogeneous. A suspension scatters in visible light and the dispersed particles settle on standing.

Q2(iii) — MAIN *$^{35}_{17}\text{Cl}$ values; atomic number & mass number definitions*

2.5/3

Correct: 17 protons, 17 electrons, 18 neutrons — all right. Mass number definition (protons + neutrons) correct.

-0.5: Atomic number defined as "number of protons and electrons" — it is the number of protons only. (First marking deducted 1 mark here in error; the slip is a minor wording error worth 0.5.)

Q2(iv) — MAIN *Sub-shell electronic configurations of Na and Cl; valence electrons*

3/3

Full marks. Na: $1s^2 2s^2 2p^6 3s^1$ (valence 1) and Cl: $1s^2 2s^2 2p^6 3s^2 3p^5$ (valence 7) both correct.

Q2(v) — MAIN *Ionization energy: definition + period and group trends*

3/3

Full marks. Correct definition with units. Period trend (increases left to right, more nuclear charge) and group trend (decreases top to bottom, more shielding) both stated and explained.

Q2(vi) — OR *MgCl₂ ionic bond formation by electron transfer*

3/3

Full marks. Mg loses 2 electrons to form Mg^{2+} ; each Cl gains 1 electron. Electron dot diagrams present; electrostatic attraction stated. Complete.

Q2(vii) — MAIN *Molecular mass of H₂SO₄*

3/3

Full marks. Calculation shown: $2(1) + 32 + 4(16) = 2 + 32 + 64 = 98$ amu. Correct.

Q2(viii) — MAIN *Balance: $Fe + O_2 \rightarrow Fe_2O_3$ with atom verification*

3/3

Full marks. Balanced equation: $4Fe + 3O_2 \rightarrow 2Fe_2O_3$. Atom count verification shown (Fe: 4=4, O: 6=6). Complete.

Q2(ix) — OR *Oxidation number of S in H₂SO₄*

3/3

A superseded false-start on the left side of the working used +2 for O (incorrect). The right-side working that followed is fully correct: $2(+1) + x + 4(-2) = 0 \rightarrow 2 + x - 8 = 0 \rightarrow x = +6$.

Benefit of the doubt: the correct final working and answer (+6) are unambiguous. Full marks. (First marking deducted 1 mark for the false-start — that was an error; the final derivation is complete and correct.)

Q2(x) — MAIN *Exothermic vs endothermic reactions; ΔH signs and examples*

3/3

Full marks. Exothermic defined (energy released, ΔH negative) with combustion as example. Endothermic defined (energy absorbed, ΔH positive) with photosynthesis as example. Both correct.

Q2(xi) — OR *Characteristics of dynamic equilibrium*

2.5/3

"Concentrations unchanged" is a valid characteristic. "Pressure/volume constant" matches the answer key's macroscopic-property requirement.

-0.5: "Temperature constant" is a condition that maintains equilibrium (a prerequisite), not a characteristic of the equilibrium state itself. Characteristics are: closed system, equal forward and reverse rates, constant concentrations of all species.

Section C — Long Answers (19/20)

Q3(i) — OR *Subatomic particles; atomic number, mass number, isotopes*

5/5

Full marks. Comparison table: proton (+1, 1 amu, nucleus), neutron (0, 1 amu, nucleus), electron (−1, ~0, shells) — all correct. Atomic number, mass number, and isotope definitions all present and accurate.

Q3(ii) — MAIN *Periodic trends: atomic size, ionisation energy, electronegativity across a period*

5/5

Full marks. All three trends across a period given with correct reasons (increasing nuclear charge, same shielding). Electropositive vs electronegative distinction made with Na/K (metals, electropositive) and Cl/O (non-metals, electronegative) as examples — the distinction is present even without the words "metals/non-metals" appearing explicitly. Benefit of the doubt: full credit.

(First marking deducted 1 mark for not writing "metals/non-metals" verbatim — that was an over-strict deduction.)

Q3(iii) — MAIN *Mole, Avogadro's number; three calculations*

5/5

Full marks. Mole defined (6.022×10^{23} particles). All three sub-calculations correct:

- Moles of 49 g H₂SO₄: $49/98 = 0.5$ mol ✓
- Molecules in 22 g CO₂: $(22/44) \times 6.022 \times 10^{23} = 3.011 \times 10^{23}$ ✓
- Mass of 2 mol NaOH: $2 \times 40 = 80$ g ✓

Critical correction on this question: Seemab answered it twice.

First attempt (scan page 7, CuO + Zn): Net ionic equation $\text{Cu}^{2+} + \text{Zn} \rightarrow \text{Cu} + \text{Zn}^{2+}$ derived correctly. Correctly identified Cu^{2+} undergoes reduction (oxidation state $+2 \rightarrow 0$) and Zn undergoes oxidation ($0 \rightarrow +2$). The redox direction is right.

Second attempt (scan page 8, redo using the correct CuSO_4 reactant): Reversed the identification — wrote Zn $0 \rightarrow -2$ and called Zn the oxidizing agent. This redo introduced errors that contradict the correct first attempt.

-1 for Part (a): The correct first-attempt derivation was never completed with explicit NAMED conclusions ("CuSO₄ is the oxidizing agent; Zn is the reducing agent"), and the contradictory redo creates ambiguity. Seemab clearly understands the direction — the gap is in naming the agents explicitly and trusting her correct first working.

Part (b) — Corrosion (2/2): Full marks. Corrosion defined correctly. Two prevention methods given: painting and galvanizing. Both correct.
(First marking treated only the page-8 redo as Seemab's answer and deducted 2 marks for "completely reversed agents" — that was a misread. The correct page-7 first attempt is her primary answer.)

Strengths

What Seemab did well

- Perfect MCQ 12/12 — 5th consecutive Chemistry (and overall) perfect-MCQ exam. Consistent recall across all 9 chapters.
- Stoichiometry Q3(iii): all three sub-calculations correct, including the molecules calculation using Avogadro's number.
- MgCl_2 ionic bond formation (2(vi) OR): electron-transfer diagram with correct charges and electrostatic attraction — complete answer.
- Sub-shell electronic configurations for Na and Cl (2(iv)): both written correctly with correct valence electron counts.
- Ionization energy (2(v)): definition plus both period and group trends with reasons — no mark lost.
- Equation balancing with atom verification (2(viii)): systematic and correct.
- Redox direction in Q3(iv) first attempt: Cu^{2+} reduced, Zn oxidized — the underlying understanding is correct.

Areas Requiring Revision

Targeted revision priorities

- **Name the agents explicitly on redox questions.** Always finish by writing "X is the oxidizing agent because it is reduced; Y is the reducing agent because it is oxidized." The understanding was there in Q3(iv) — it just was not stated. (Q3(iv))
- **Trust a correct first derivation.** If the first attempt gives the right answer, do not overwrite it with a redo. The redo on Q3(iv) introduced errors that contradicted the correct page-7 working. Cross out the redo, not the original.
- **Use the exact reactant given in the question.** The question specified CuSO_4 ; the first attempt used CuO. (Q3(iv))
- **Atomic number = protons only.** Electrons equal protons in a neutral atom, but the definition is "number of protons." Writing "protons and electrons" is wrong. (Q2(iii))
- **A suspension is heterogeneous.** Its particles are large enough to be seen and settle on standing — it does not form a uniform mixture. (Q2(ii) OR)
- **Characteristics of dynamic equilibrium vs conditions.** Characteristics: closed system, equal forward/reverse rates, constant concentrations. Constant temperature is a condition that maintains equilibrium, not a feature of the equilibrium state itself. (Q2(xi) OR)

Chemistry Exam History (all 5 attempts)

Exam	Chapters	Date	MCQ	Total	%	Grade
Exam 01	Ch 1-2	2026-04-12	12/12	64/65	98.5%	A+
Exam 02	Ch 3-5	2026-04-17	12/12	59.5/65	91.5%	A+
Exam 03	Ch 6	2026-05-12	12/12	57.5/65	88.5%	A+
Exam 04	Ch 7	2026-06-12	12/12	60/65	92.3%	A+
Exam 12 ★	Ch 1-9	2026-06-24	12/12	62.5/65	96.2%	A+

Seemab's 2nd-best Chemistry result (96.2%), behind only Exam 01 (98.5%), and her highest-scoring paper on the full Ch 1-9 syllabus.